

TRIPOLI INTL, Libyan Arab Jamahiriya

WMO: 620100

Lat: 32.664N

Lon: 13.159E

Elev: 80

StdP: 100.37

Time Zone: 2.00 (E02)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.5	5.8	-4.2	2.7	18.1	-2.1	3.2	16.9	11.5	14.6	10.3	15.6	2.4	210	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	42.1	22.6	40.0	22.3	38.0	21.9	26.8	35.2	25.6	32.9	24.6	31.7	5.1	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	20.1	30.2	23.7	18.7	29.3	22.8	17.7	28.5	84.4	35.1	78.8	32.7	75.0	31.5	33.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	9.5	8.4	DB	1.7	45.5	1.5	1.3	0.7	46.5	-0.2	47.3	-1.1	48.0	-2.1	49.0	(4)
			WB	0.4	28.6	1.3	2.8	-0.6	30.7	-1.3	32.3	-2.1	33.9	-3.0	36.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.3	12.9	13.2	16.3	19.8	23.8	27.2	28.8	29.7	27.8	23.9	18.3	14.0
	DBStd	6.91	2.25	3.01	3.76	4.08	4.20	3.93	3.08	2.77	3.04	3.22	3.13	2.64
	HDD10.0	7	2	4	0	0	0	0	0	0	0	0	0	1
	HDD18.3	614	169	148	88	28	2	0	0	0	0	1	39	139
	CDD10.0	4144	92	93	194	293	428	515	583	609	533	430	248	124
	CDD18.3	1711	1	4	24	71	173	265	324	351	283	172	37	4
	CDH23.3	19956	13	61	298	792	1942	3223	4158	4537	3121	1486	282	43
	CDH26.7	10790	2	17	118	362	1027	1856	2439	2646	1593	644	76	10
Wind	WSAvg	3.7	3.3	3.6	3.8	4.3	4.4	4.3	3.6	3.6	3.7	3.3	3.2	3.4
Precipitation	PrecAvg	275	58	30	26	17	3	1	0	0	11	34	43	54
	PrecMax	525	173	83	100	105	26	18	4	3	80	136	172	212
	PrecMin	49	2	0	0	0	0	0	0	0	0	0	0	2
	PrecStd	104	43	24	26	23	6	3	1	1	17	38	41	53
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.0	29.0	34.7	37.8	41.9	44.6	44.1	43.1	41.5	37.8	31.7	27.8
		MCWB	13.2	14.8	17.9	20.1	21.7	24.0	22.7	23.0	23.1	21.1	17.9	14.3
	2%	DB	21.9	24.8	29.9	33.9	38.2	41.3	41.4	41.0	38.9	34.9	28.9	23.8
		MCWB	12.8	14.0	16.7	17.7	20.9	21.7	22.6	23.4	21.5	20.3	17.0	14.7
	5%	DB	19.8	21.9	26.2	30.9	35.3	38.3	38.8	38.9	36.2	32.8	26.8	21.1
		MCWB	12.6	13.1	15.1	17.6	19.9	21.9	22.3	22.8	21.8	19.6	16.8	13.5
	10%	DB	18.0	19.2	23.7	28.1	32.8	35.6	36.2	36.8	34.1	30.2	24.6	19.2
		MCWB	11.9	12.5	14.6	16.5	19.0	20.9	21.6	22.5	21.8	19.7	16.2	13.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.7	17.4	20.9	24.1	26.3	28.9	27.3	28.5	27.4	25.1	21.3	18.0
		MCDB	20.4	23.5	28.3	31.0	37.2	41.5	37.5	37.3	34.2	31.4	27.2	22.2
	2%	WB	14.2	15.2	18.0	20.4	23.4	25.6	25.4	26.7	25.7	23.2	19.6	16.0
		MCDB	18.8	21.9	25.8	29.1	33.9	36.2	33.8	34.0	31.5	29.0	24.7	20.5
	5%	WB	13.4	14.0	16.4	18.6	21.4	23.5	24.4	25.5	24.7	22.2	18.4	14.9
		MCDB	17.6	19.9	23.4	27.4	30.9	32.2	32.4	32.2	30.3	28.1	23.6	19.0
	10%	WB	12.7	13.0	15.3	17.2	20.1	22.3	23.6	24.6	23.8	21.2	17.4	13.9
		MCDB	16.8	17.9	21.6	25.5	28.8	30.0	31.1	31.3	29.7	27.0	22.3	18.0
Mean Daily Temperature Range	5% DB	MDBR	10.1	11.2	12.5	12.9	13.3	13.4	13.3	12.9	11.7	11.5	11.0	10.0
		MCDBR	13.2	15.8	16.7	17.3	16.8	17.0	16.5	15.7	14.6	14.6	13.6	13.2
		MCWBR	7.2	7.9	7.6	7.2	6.5	6.6	6.4	5.9	5.4	5.9	6.2	7.2
	5% WB	MCDBR	10.9	13.5	14.5	15.2	14.9	14.2	13.4	12.9	11.9	12.1	11.7	10.7
	MCWBR	6.9	8.0	7.8	7.8	7.3	6.8	6.3	6.1	5.5	6.0	6.6	7.0	
Clear-Sky Solar Irradiance	taub	0.380	0.404	0.481	0.537	0.539	0.485	0.486	0.484	0.515	0.472	0.420	0.388	
	taud	2.220	2.137	1.922	1.814	1.836	1.992	2.004	2.024	1.947	2.025	2.118	2.196	
	Ebn at Noon	817	836	794	769	771	811	807	803	756	754	761	781	
	Edh at Noon	114	137	183	213	210	179	176	170	177	152	126	111	
All-Sky Solar Radiation	RadAvg	3.10	3.96	5.24	6.35	7.15	7.80	7.91	7.12	5.75	4.33	3.33	2.70	
	RadStd	0.19	0.22	0.30	0.35	0.30	0.30	0.14	0.19	0.23	0.27	0.15	0.20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page